

Do People Really Want Humanoid Robots in Their Homes?

Companies pour billions into a vision we should be skeptical of

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Screenshots of marketing videos of three humanoid robots specifically advertised for in-home

<https://spectrum.ieee.org/home-humanoid-robots-survey>

use, shown performing a range of tasks. Neo by 1X Technologies [top], Figure 02 by Figure [middle], and Optimus by Tesla [bottom].

1X Technologies; Figure; Tesla
[humanoid-robots](#) [guest-article](#) [home-robots](#) [robotics](#)

I've been teaching [robotics](#) at the University of Washington for more than a decade. Every class begins with “robotics news of the week.” For years, [humanoid robots](#) appeared only occasionally—usually in the form of viral clips of Boston Dynamics’ Atlas doing [parkour](#) or [RoboCup](#) humanoid league bloopers that served more as comic relief than serious news.

But over the past few years, things have shifted. Each week brings another humanoid demo, each flashier than the last, as companies race to showcase new capabilities. And behind those slick videos lies a flood of [venture capital](#). Humanoid robotics has become a billion-dollar frenzy.

The scale of investment is astonishing. Just a year ago, [Figure AI’s \\$2.6 billion valuation](#) seemed extraordinary—until its latest funding round catapulted it to [\\$39.5 billion](#). Investors large and small are rushing in, and tech giants like [Microsoft](#), [Amazon](#), [OpenAI](#), and [Nvidia](#) are scrambling to get a foothold for fear of missing out. [Tesla](#) is [pouring resources into its Optimus robot](#), while [China](#) has committed more than [\\$10 billion in government funding](#) to drive down costs and seize market dominance. [Goldman Sachs](#) now projects the global humanoid market could reach [\\$38 billion by 2035](#).

This surge of interest reflects a long-standing dream in robotics: If machines could match human form and function, they could simply step into human jobs without requiring us to change our environments. If humanoids could do everything people can, then in theory they could replace workers on the factory floor or in warehouse aisles. It’s no surprise, then, that many humanoid companies are targeting what they believe are sectors with labor shortages and undesirable jobs—[manufacturing](#), [logistics](#), [distribution](#), [retail](#)—as near-term markets.

Factories first, homes next?

A subset of humanoid companies see homes as the next frontier. [Figure AI claims](#) humanoids will revolutionize “assisting individuals in the home” and “caring for the elderly.” Its marketing materials show robots handing an apple to a human, making coffee, putting away groceries and dishes, pouring drinks, and watering plants. [Tesla’s Optimus](#), similarly branded as an “autonomous assistant, humanoid friend,” is shown folding clothes, cracking eggs, unloading groceries, receiving packages, and even playing family games. The [Neo humanoid by 1X Technologies](#) appears targeted solely at in-home use, with the company declaring that “1X bets on the home” and is “building a world where we do more of what we love, while our humanoid companions handle the rest.” Neo is depicted vacuuming, serving tea, wiping windows and tables, and carrying laundry and grocery bags.

All these glossy marketing videos struck a personal chord with me. I have always dreamed of robots in homes—and I know [I am not alone](#). Like many roboticists of my generation, my

earliest memories of robots trace back to Rosie the Robot from [The Jetsons](#). I dedicated my career to getting assistive robots into homes. In 2014, my students and I placed a [PR2 robot](#) in a home in Arizona, where it failed miserably at most tasks—though we learned a great deal in the process. Later, I was part of more successful in-home deployments of a [Stretch robot](#) and an [assistive feeding robot](#). I even found myself enjoying housework because it gave me a chance to [analyze the tasks it entailed](#) with an eye toward someday automating them. For years, I promoted my work under a personal motto: “I want robots to do all the chores by the time I retire,” often joking that I might never retire.

Yet when billion-dollar companies began chasing the same dream, I found myself reacting with unease. I had always imagined that [home robots](#) would be more like Rosie—robotic and cartoonish—and my own research moved further and further away from the human form because non-humanoid robots were more practical and preferred by users. I struggled to picture a humanoid in my own house—or any of the homes where I had deployed robots. And after years of human-centered research in robotics, I could not imagine users welcoming humanoids into their homes without hesitation. Still, I assumed someone must want them. Surely some fraction of those billions had gone into market research and customer insight. And I wanted to know what they knew.



[Left] Three real-world humanoids shown to participants in our study (Figure, Optimus, Neo). [Right] Examples of three general-purpose robots with few or no humanlike features ([PR2](#), Fetch, Stretch). Maya Cakmak

What people actually think

To find out, my students and I set out to better understand what the public thinks about humanoid robots in the home. We surveyed 76 participants from the U.S. and the U.K., asking whether they considered humanoids in the home acceptable, which designs they preferred, and why. We also presented them with imagined scenarios where either a humanoid or a special-purpose robot assisted an older adult with tasks like eating, dressing, or vacuuming, and asked which they would choose. The results are detailed in our paper “[Attitudes Towards Humanoid Robots for In-Home Assistance](#),” presented this week at the [IEEE International Conference on Robot and Human Interactive Communication](#) (RO-MAN).

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Our survey showed that people generally prefer special-purpose robots over humanoids. They see special-purpose robots as safer, more private, and ultimately more comfortable to have in their homes and around loved ones. So, while humanoid companies (and their investors) dream of a single humanoid capable of doing it all, our survey participants seem to be more on board with a toolbox of smaller, specialized machines for most tasks: a [Roomba](#) for cleaning, a medication dispenser for pills, a stairlift for stairs.

Nevertheless, most survey participants considered humanoids in the home acceptable. Some even preferred humanoids for certain tasks, especially when the special-purpose alternative was more speculative—like a dressing assistant robot. When shown images of Neo, Figure 02, and Optimus performing [household](#) tasks, they agreed the robots looked useful and well-suited for homes. Many said they would feel comfortable having one in their own home—or in the home of a loved one. Of course, we had framed the scenarios optimistically: Participants were told to assume the robots had passed extensive safety testing, were approved by regulators, and would be covered by insurance—assumptions that may be decades away from reality. And we can safely assume that finding humanoids “acceptable” doesn’t mean people actually want them—or that they’d be willing to pay for one.



AI-generated images of humanoid and special-purpose robots across eight tasks used in our questionnaires. Cakmak et al, 2025

Are home humanoids safe?

Unsurprisingly, the task context impacted whether people were open to humanoids in the home. Participants balked at imaginary scenarios involving safety-critical assistance—such as being carried down a staircase—responding with visceral rejections like “absolutely not in a million years.” Whereas for tasks that require little interaction—such as folding laundry—most were willing to imagine a humanoid lending a hand.

Even with our reassurances about safety, people readily imagined hazards: Humanoids could trip, stumble, or tip over; they might [glitch](#), run out of battery, or malfunction. The idea of a robot handling hot surfaces or sharp objects was also mentioned by multiple participants as a serious concern.

Privacy was another major concern. Participants worried about camera data being sent to the cloud or robots being remotely controlled by strangers. Several pointed out the security risks—any Internet-connected device, they noted, could be hacked.

Even participants who saw clear benefits often described a lingering unease. Several described the robots as “creepy” or “unsettling,” and a few explicitly mentioned the [uncanny valley effect](#), pointing in particular to the black face [masks](#) common on this new generation of humanoids. One participant described the masks as creating an “eerie sensation, the idea that something might be watching you.” I felt a similar conflict watching a video of Neo (1X) [sitting on a couch after finishing its chores](#)—a scene that was meant to be comforting but instead left me unsettled.

A common reason participants preferred special-purpose robots was space. Humanoids were described as “bulky” and “unnecessary,” while specialized robots were seen as “less intrusive” and “more discreet.” These comments reminded me of user research conducted in [Japan](#) by the [Toyota Research Institute](#), which led to a [ceiling-mounted robot design](#) after finding that limited floor space was a major barrier to adoption. The same thought struck me at home when I showed an in-home humanoid video to my 9-year-old and asked if we should get one. He replied: “But we don’t have an extra bed.” His answer nailed the point: If your home doesn’t have room for another human, it probably doesn’t have room for a humanoid.

Very big ifs

In the end, the study didn’t fully answer my question about what these companies know that I don’t. Participants said they would accept humanoids—if they were safe, worked reliably, and didn’t cost more than the alternatives. Those are very big ifs.

And of course, our study asked people to use their imaginations. Looking at a picture is not the same as sharing your living room with a six-foot metal figure that moves—in reality, their reactions might be very different. Likewise, picturing yourself someday needing help with eating, dressing, or walking is very different from already relying on that help every day. Perhaps for those already living with these needs, the immediacy of their situation would make the promise of humanoids more compelling.

To probe further, I asked the same question at the [HRI 2025 Physical Caregiving Robots Workshop](#) to a panel of six people with motor limitations who are experienced users of assistive robots. Not one of them wanted a humanoid. Their concerns ranged from “it’s creepy” to “it has to be 100 percent safe because I cannot escape it.” One panelist summed it up perfectly: “Trying to make assistive robots with humanoids would be like trying to make [autonomous cars](#) by putting humanoids in the driver’s seat and asking them to drive like a human.” After all, it was obvious to investors that the better path to [autonomous vehicles](#) was to modify or redesign vehicles for autonomy, rather than replicate human drivers. So why are they convinced that replicating humans is the right solution for the home?

What’s the alternative?

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Special-purpose robots may be preferable to humanoids, but building a dedicated machine for every possible task is unrealistic. Homes involve a [long tail](#) of chores, and general-purpose robots could indeed provide enormous value. However, the humanoid form is likely overkill, since much simpler designs—such as [wheeled robots](#) with basic pinch grippers—can already accomplish a great deal and are far more attainable. And people will likely accept modest [changes to their homes](#) to expand what these robots can do, just as [Roomba owners move furniture](#) to let their vacuums work. After all, our homes have already transformed around new technologies—cars, [appliances](#), televisions—so why not for robots, if they prove just as valuable?

But beyond the unnecessary complexity, a more important issue about the humanoid form may be that users find it less desirable than simpler alternatives. Research has long shown that highly [humanlike robots can trigger negative emotional responses](#), and our study suggests that is true of the latest generation of humanoids. Simpler designs with more [cartoonlike features are more likely to be accepted](#) as companions. We may even want home robots with no humanlike features at all, so they can be viewed as tools rather than social agents. I believe those who would benefit most from in-home robots—including the [rapidly growing population of older adults](#)—would prefer robots that empower them to do things for themselves, rather than ones that attempt to replace human caregivers. Yet humanoid companies are openly pursuing the latter.

Only time will tell whether humanoid companies can deliver on their promises—and whether people, myself included, will welcome them into their homes. I hope our findings encourage these companies to dig deeper and share their insights about in-home humanoid customers. I'd also like to see more capital directed toward alternative robot designs for the home. In the meantime, my students and I can't wait to get our hands on one of these humanoids—purely in the name of science—bring it to older adults in our communities, and hear their unfiltered reactions. I can already imagine someone saying, “It better not sit in my recliner when I'm not looking,” or, “If it's going to live here, it better pay rent.”